

DIABETIC RETINOPATHY SCREENING REPORT

Explainable AI Screening Prototype

Patient ID: 001
Name: shafin
Age: 19 Sex: Male Eye: Left Eye
Report Number: DRX-20260909-202746-18CE

Input Fundus Image



prototype_fundus.png

1. Image Quality Assessment

IMAGE QUALITY: ACCEPTABLE
Focus: 0.0009
Brightness: 0.3938
Contrast: 0.0597
FOV Ratio: 0.7918
Illumination: 0.0677

2. DR Classification

Predicted Grade: Grade 0
Severity: No DR
Model Confidence: 99.97%

3. Class Probabilities

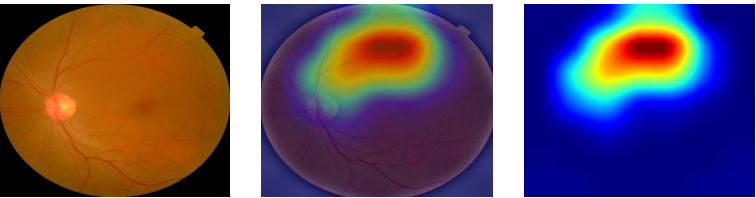
No DR: 99.97%
Mild DR: 0.02%
Moderate DR: 0.00%
Severe DR: 0.00%
Proliferative DR: 0.01%

4. Referable DR Assessment

Referable Probability: 0.01%
Decision Threshold: 35%
Decision: NON-REFERABLE
NON-REFERABLE DR
Routine monitoring recommended.

5. Explainability - Grad-CAM

Grad-CAM highlights image regions that contributed to the predicted DR class. The visualization provides model-level evidence and is not a definitive lesion diagnosis



reprocessed Fundu Grad-CAM Overlay Attention Heatmap

6. MATLAB Preprocessing

- ✓ Retinal field detection and cropping
- ✓ Illumination correction
- ✓ CLAHE enhancement
- ✓ Mild denoising
- ✓ Resizing to 224 x 224

7. Screening Summary

Predicted Grade: Grade 0

Severity: No DR

Model Confidence: 99.97%

Referable Probability: 0.01%

Decision: NON-REFERABLE

Recommendation: Routine monitoring recommended.

Research Prototype - Not a Final Clinical Diagnosis

Analysis completed successfully.